

# Using simulation to determine the need for ICU beds for surgery patients

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**Background.** As the need for surgical ICU beds at the hospital increases, the mismatch between demand and supply for those beds has led to the need to understand the drivers of ICU performance.

**Method.** A Monte Carlo simulation study of ICU performance was performed using a discrete event model that captured the events, timing, and logic of ICU patient arrivals and bed stays.

**Results.** The study found that functional ICU capacity, ie, the number of occupied ICU beds at which operative procedures were canceled if they were known to require an ICU stay, was the main determinant of the wait, the number performed, and the number of cancellations of operative procedures known to require an ICU stay. The study also found that actual and functional ICU capacity jointly explained ICU utilization and the mean number of patients that should have been in the ICU that were parked elsewhere.

**Conclusion.** The study demonstrated the necessity of considering actual and functional ICU capacity when analyzing surgical ICU bed requirements, and suggested the need for additional research on synchronizing demand with supply. The study also reinforced the authors' sense that simulation facilitates the evaluation of trade-offs between surgical management alternatives proposed by experts and the identification of unexpected drawbacks or opportunities of those proposals. (Surgery 2009;146:608-20.)

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THE JEWISH GENERAL HOSPITAL provides a broad range of inpatient and outpatient services to a large and diverse population in and around Montreal. It has 637 beds, including 154 surgical beds and a combined Medical-Surgical Intensive Care Unit (ICU) that typically has between 14 and 16 staffed beds. Demand for ICU beds comes from the emergency department, from emergent, urgent and elective operative procedures, and from the medical wards. This demand has increased due to the hospital's emergency department becoming the second busiest in the province, because the number of postoperative patients requiring ICU stays per year has increased, and because the hospital has become a referral center for nearby facilities. While the likelihood of different levels of demand on a particular day can be determined from historical data, actual demand for ICU beds on a particular day is random. This is because the number of new patients from the

emergency department and the medical wards requiring an ICU stay on that day, and the length of stay of patients already in the ICU, are not known in advance. From April 1, 2007, through March 31, 2008, of approximately 5,475 ICU bed days available to medical and surgical patients, there were 780 ICU bed stays totaling 2,934 days by 741 surgical patients. There were also 173 cancellations of operative procedures due to the unavailability of ICU beds, and a significant amount of unmeasured time during which surgical patients who should have been sent to the ICU were sent elsewhere.

Despite the increased demand for critical care, the hospital has not significantly expanded its critical care areas. The resulting mismatch between demand and supply of ICU beds led to cancellation of procedures known to require an ICU stay, which in turn produced longer average wait times for patients requiring those procedures. It also led to patients not being admitted to the ICU when they should have been. These consequences created a situation in which access to ICU beds became a critical concern for the hospital; in December 2007 there was a period of 1 week in which all elective procedures known to require an ICU stay were canceled. To address this, plans are being developed to add critical care beds to the hospital.

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As part of the planning process, it was deemed necessary to understand the mechanisms available for reducing these problems. The most obvious mechanism would be to increase the total number of ICU beds. The disadvantages of this mechanism include the expense of adding and staffing ICU beds, as well as the difficulty of hiring and retaining ICU nurses. A related mechanism, used regularly by hospital interventionists, would be to control the functional ICU capacity, ie, the number of occupied ICU beds at which scheduled procedures known to require an ICU stay were canceled. The disadvantages of this mechanism are the same as those of the first mechanism, as significant increases in functional ICU capacity generally require similar increases in actual ICU capacity. A third mechanism would be to adjust instantaneous demand to more closely match instantaneous supply.<sup>1,2</sup> One possible way of doing this is to limit the number of procedures known to require ICU stays performed each day.<sup>1</sup> A second possible way of doing this is to delay the scheduling of procedures for a range of dates to a date close to the start of that range, so that ICU utilization as of that date can be used to inform the scheduling process. A fourth mechanism would be to reserve a set of ICU beds for patients undergoing elective procedures. While this would reduce cancellations, it would also increase the time that nonsurgical patients wait for access to the ICU.<sup>3,4</sup>

Once having identified actionable mechanisms for improving ICU performance, it was deemed necessary to analyze their impact on ICU performance. One possible tool for performing this analysis was a mathematical theory of waiting based on probability called queuing theory. This theory was originally developed at the turn of the 20th century to determine the number of switches required in phone exchanges to minimize delays in connecting customer calls. A major advantage of using queuing theory is that it provides mathematical formulas for computing performance measurements. However, the use of queuing theory was not, in the authors' opinions, appropriate for this analysis. The reason for this is that when dealing with a process having several activities, such as an operative procedure, followed by an ICU stay, followed by a stay in the surgical ward, possibly followed by another ICU stay, the queuing models that are most tractable for this analysis are Jackson queuing network models.<sup>5</sup> One of the requirements for using Jackson queuing models is that the probability distribution of each activity's time be independent and identically exponentially distributed. An implication of this requirement is

that if the average time to perform all procedures is four hours, then the average length of stay in the ICU, and the average length of stay after that in a surgical ward, would both also be four hours. Since there was no reason to believe that this requirement would be satisfied, use of queuing models did not seem appropriate.

Thus for this research, Monte Carlo simulation was used. Monte Carlo simulation can be thought of as being a statistical experimentation technique used to calculate the probability or expected value of certain results, such as the number of procedure cancellations. In Monte Carlo simulation, a model of a phenomenon is repeatedly run using different sets of random or pseudo-random data, much like airplane models are tested in wind tunnels with different atmospheric conditions. When completed, the results of the individual runs, which are statistically independent of each other because they were run with different sets of random or pseudo-random data, are statistically analyzed in a manner similar to the statistical analysis performed in randomized pharmaceutical studies. The reason for using pseudo-random data is that when simulation results may need to be replicated, random data is usually generated algorithmically. Since it is impossible to generate truly random data algorithmically, a considerable amount of research<sup>6</sup> has gone into creating pseudo-random number generators that generate numbers that statistically appear random.

To conduct the simulation study, a discrete event model of the processes underlying ICU bed usage was built to measure the effect on performance of the mechanisms discussed above. Discrete event models are ideal for analyzing ICU bed requirements, as they make it possible to simulate the timing and occurrence of the events that drive both ICU usage and performance. These events include admission request submissions, ICU stay starts, ICU stay ends, patient arrivals to the ICU, patient departures from the ICU, bed preparation starts, and bed preparation completions. Discrete event models also make it possible to model entities, eg, operating rooms, surgeons, and patients, and to calculate performance statistics. Discrete event simulation models are also used by other researchers<sup>1,2,7-10</sup> for modeling ICU or surgical bed requirements.

The goal of the simulation study was to identify potential avenues for further improving ICU performance, in the context of a surgical-patient-only ICU. The first step to meeting that goal was to develop a simulation model that could be used for this purpose. The second step was to use the

**Table I.** Required probabilities, probability distributions, and other data

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|                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The probability distribution of the time between requests for operative procedures.                                                                                                                                               |
| The probability of a randomly selected request for an operative procedure being a particular type of operative procedure, for each type of operative procedure.                                                                   |
| The probability of an operative procedure being an emergency procedure, for each type of operative procedure.                                                                                                                     |
| The probability distribution of time in the OR for an operative procedure, for each type of operative procedure.                                                                                                                  |
| The probability of requiring an ICU stay immediately after an operative procedure, for each type of operative procedure.                                                                                                          |
| The probability distribution of the time between leaving the OR to when an ICU stay started, given that there was an ICU stay that did not start immediately after the operative procedure, for each type of operative procedure. |
| The probability distribution of the time in a first ICU visit stay given that there is a first ICU stay, for each type of operative procedure.                                                                                    |
| The probability of needing a second ICU stay, given that there is a first ICU stay, for each type of operative procedure.                                                                                                         |
| The probability distribution of the time from leaving a first ICU stay to when a second ICU stay started, given that there were 2 ICU stays, for each type of operative procedure.                                                |
| The probability distribution of the time of a second ICU visit stay, given that there is a second ICU stay, for each type of operative procedure.                                                                                 |
| The surgical service corresponding to each type of operative procedure; operative procedures that were performed by more than 1 service were treated as different types of operative procedures.                                  |

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simulation model to determine the relationships between ICU performance and the first 2 of the mechanisms discussed above.

## METHODS

To perform the study, the first author spent approximately 125 hours performing the steps explained in the following subsections; additional unmeasured time was also spent by the chief of surgical services, the chief of surgical nursing, members of the ICU bed shortage task force (listed below), and members of the hospital's information management team.

### **Determining the underlying logic of the model.**

To determine the underlying logic of the processes to be modeled, meetings were conducted with the chief of surgical services, the surgical nursing director, and physicians from the ICU. During these meetings, the processes involved in ICU stays, and the interactions between those processes, were discussed in detail. The first author also visited and observed relevant areas of the hospital.

**Determining needed probabilities, probability distributions and parameters.** The probabilities, probability distributions, and parameters needed for the study are listed in [Table I](#). By probability we mean the relative likelihood of a single event occurring, eg, the relative likelihood of getting a head on a single throw of a fair coin is  $\frac{1}{2}$ . By probability distribution we mean the relative likelihood of each possible event occurring, eg, the relative likelihood of getting a 1, 2, 3, 4, 5, and a 6 on a single throw of a fair die is  $\frac{1}{6}$ ,  $\frac{1}{6}$ ,  $\frac{1}{6}$ ,  $\frac{1}{6}$ ,  $\frac{1}{6}$ , and  $\frac{1}{6}$ . By cumulative probability distribution,

for the throw of a single fair die, we mean the probability of getting a 1 is  $\frac{1}{6}$ , the probability of getting a 1 or a 2 is  $\frac{2}{6}$ , and so on.

**Determining data sources.** The data sources used to estimate the probabilities and probability distributions are listed in [Table II](#). All but one of the data sources were from hospital databases. The data that was not from a hospital database was provided by the hospital's chief anesthetist and an intensivist, based on their personal experience.

**Estimating probabilities and probability distributions.** To estimate the probabilities of particular events needed for the simulation, the number of times the particular event occurred in the collected data was divided by the total number of relevant events in the collected data. For example, to compute the probability of an inguinal hernia repair procedure being an emergency operative procedure, the number of emergency inguinal hernia repair procedures was divided by the total number of inguinal hernia repair procedures.

For all distributions except that of the time between requests for operative procedures, cumulative probability distributions were estimated by interpolating the values corresponding to the 1st, 2nd, 3rd, . . . and 100th percentiles from the collected data. Thus, if 5 data points were collected, the 0th percentile was set to the smallest value, the 25th percentile was set to the second smallest value, the 50th percentile was set to the third smallest value, and so on. After this was done, the 1st through 24th percentiles were estimated by interpolating between the 0th and 25th percentiles, the 26th through 49th percentiles were

**Table II.** Source datasets

| <i>Description</i>                                                                                                     | <i>Data source</i>                                                                | <i>Fields</i>                                                | <i>Field type</i> |
|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------|
| A list of all surgeries performed by each service from April 2007–March 2008.                                          | This data was obtained from the hospital's surgical information system.           | Patient identifier                                           | Numeric           |
|                                                                                                                        |                                                                                   | Intervention type                                            | Coded as a number |
|                                                                                                                        |                                                                                   | Elective/urgent indicator                                    | Coded             |
|                                                                                                                        |                                                                                   | Minute of arrival to OR relative to start of data collection | Numeric           |
|                                                                                                                        |                                                                                   | Minutes spent in OR                                          | Numeric           |
| The probability of requiring an ICU stay immediately after the intervention, for each type of intervention.            | This data was provided by the chief anesthetist, and an intensivist from the ICU. | Surgeon                                                      | Coded as a number |
|                                                                                                                        |                                                                                   | Intervention type                                            | Coded as a number |
| A list of all ICU visits by surgical patients, the time at which their ICU stay started, and the length of their stay. | This data was provided by two intensivists from the ICU.                          | ICU stay probability                                         | Numeric           |
|                                                                                                                        |                                                                                   | Patient identifier                                           | Coded as a number |
|                                                                                                                        |                                                                                   | ICU stay start minute relative to start of data collection   | Numeric           |
| The surgery schedule for each service.                                                                                 | This data was obtained from the master surgery schedule.                          | ICU stay minutes                                             | Numeric           |
|                                                                                                                        |                                                                                   | Service identifier                                           | Coded as a number |
|                                                                                                                        |                                                                                   | Block start minute relative to start of data collection      | Numeric           |
|                                                                                                                        |                                                                                   | Block minutes                                                | Numeric           |

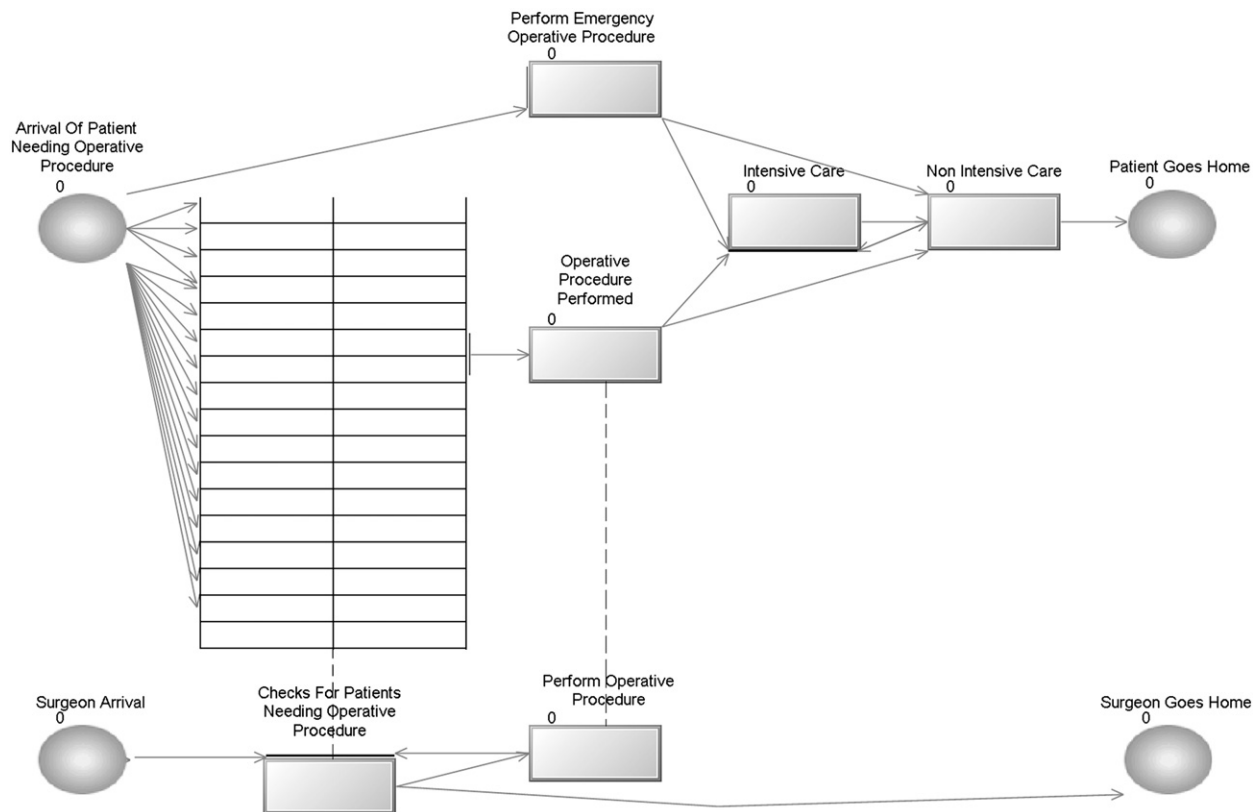
estimated by interpolating between the 25th and 50th percentiles, and so on. This approach was used because of the impracticality of fitting distributions and testing the goodness of their fit, for the large number of distributions that were required, which included the distribution of intervention time and ICU stay time for each type of intervention. Because there were no a priori reasons for trying to fit the data to particular probability distributions, there was no reason to believe that the quality of the estimated distributions were of any less quality than what might be achieved, had each distribution been individually fitted to an existing probability distribution.

Since requests for operative procedures were not entered into the surgical information system by requesting surgeons, the actual times at which requests were created was unknown. Since the likelihood of a new need for an operative procedure occurring in a particular moment seemed to be independent of past history in large patient populations, it was hypothesized that the distribution of time between new needs for operative procedures was exponentially distributed, because the exponential distribution is the only continuous probability distribution having this property of independence. It was also observed that the discrepancies between a plot of actual times between requests, and times between requests predicted by the exponential distribution, could be explained by the fact that requests tended to be entered by administrative staff in batches. This suggested that

the actual time between new requests for operative procedures, as opposed to the time between the entry of the requests, was exponentially distributed. It was also observed that there was almost always a significant queue of operative procedures for each service, so that not using the exact time at which requests for operative procedures were submitted would not significantly affect simulation results. Therefore, it was decided to use the exponential distribution, with a mean interarrival time equal to the length of the data collection period divided by the total number of requests for operative procedures, as the distribution describing the times between requests for those procedures.

#### **Building the discrete event simulation model.**

The discrete event simulation model was built to reflect patient and surgeon flows; the visual part of the simulation model is displayed in the [Figure](#). The flow of patients is modeled in the top part of the drawing, starting with the generation of operative procedure requests in the top left icon. Using analyst specified logic, the simulation model randomly specifies, using the generated probabilities and probability distributions discussed above, whether the procedure is elective or urgent, the number of minutes the procedure will take, whether the patient will require 0, 1, or 2 ICU stays, the number of minutes until each ICU stay starts, and the number of minutes each ICU stay lasts. If the procedure is urgent, it is performed immediately, otherwise patients queue up by service into normal and high priority queues (displayed



**Figure.** Visual representation of simulation model.

to the left of the Surgery Performed activity). For the purpose of this study, high priority queues were reserved for procedures known to require ICU bed stays.

In the bottom part of the model, surgeons arrive to perform procedures according to the actual, predetermined, and nonrandom schedule. The simulation then uses analyst specified logic to determine subsequent surgeon behavior. In particular, if there is a high priority procedure to be performed for their service, and the number of occupied beds will be less than the functional ICU capacity within 4 hours after the start of the procedure, they perform that procedure first, regardless of the amount of time it will take; the four hour period is used to allow for ICU beds that become available after the start of the procedure. They then search within their service's high priority queue, and then their service's low priority queue, for other procedures that fit into the time remaining on that day. When they can perform no more procedures on that day, surgeons go home.

When a surgeon finds a procedure to perform, analyst specified logic, visually indicated by the leftmost dashed vertical line, specifies that the patient upon whom the procedure is to be

performed is to be moved into the operating room. The surgeon then performs the procedure on the patient. After the procedure is completed, the patient is moved either to the ICU, or to the Post Anesthesia Care Unit (PACU) followed by a surgical ward, depending on whether the patient has been identified as requiring an ICU stay immediately after surgery. When patients have adequately recovered, they are released. As it was not necessary for this study, the model does not differentiate between time spent in the PACU and time spent in surgical wards.

In addition to that mentioned above, additional analyst specified logic is used to load derived probability distributions into the simulation platform, generate pseudo-random quantities from those distributions, and compute and save the performance measures discussed above. Analyst specified logic is also used to run the model 64 times for each level of functional ICU capacity, and to compute trial statistics on the performance measures listed in Table III computed in each run.

To generate a pseudo-random value from an estimated distribution, analyst specified logic randomly picks a percentile between 0 and 100 using the simulation software's uniformly distributed



**Table III.** Performance measures analyzed in the study

| <i>Performance measure</i>                                                                                            | <i>Explanation</i>                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The average number of days patients wait for cardiac interventions known to require an ICU stay.                      | Since the vast majority of all interventions requiring immediate ICU stays were cardiac operative procedures, the change in wait time of patients waiting for cardiac interventions requiring an immediate ICU stay was used as a proxy for determining the magnitude of change in wait time, for all patients, that would occur when changing functional ICU capacity. |
| The number of cardiac operative procedures, known to require an ICU stay, that were actually performed.               | This performance measure was collected to demonstrate that when functional ICU capacity is insufficient that the average wait performance measure does not include patients that never actually received the operative procedure during the simulation.                                                                                                                 |
| The average number of patients that should have been in the ICU, but were instead parked elsewhere.                   | Patient parking is generally considered to be an activity that should be minimized.                                                                                                                                                                                                                                                                                     |
| The number of blocks in which at least one intervention was canceled because of insufficient functional ICU capacity. | The mechanism used for scheduling interventions in the simulation model would have led to an overcount of the number of interventions canceled. As a result, this metric was used instead.                                                                                                                                                                              |
| Utilization of ICU beds                                                                                               | Because low utilization of ICU beds is very expensive, it is desirable that this metric, and the number of interventions performed, be fairly high, while minimizing the other metrics.                                                                                                                                                                                 |

pseudo-random number generator. Analyst specified logic then selects the corresponding value from the cumulative probability distribution.

**Verifying the model.** To verify the model, the following steps were performed. First, the number of queued up operative procedure requests at the end of the first 3-month warmup period was compared to that expected in a 3-month period. Second, the number of operative procedures performed each year was compared to that expected. Third, the number of queued up operative procedure requests at the end of the simulation run was compared to that expected on an ongoing basis.

After concluding that the results of the comparisons were reasonable, the model was presented to a committee of subject matter experts, a sitting hospital task force deliberating over how to address ICU bed shortages, to ensure that it met their criterion of reasonableness. The presentation included an animation of the model so that the task force could see and understand its flow, and discussions of the time frame for which the model was run, how patients were queued and subsequently selected for procedures, the number of times the model was run, and the data extracted from the model. The task force included the chief of surgical services, the chief of nursing, the critical care nursing director, the surgical nursing director, the ICU head nurse, and the ICU chief.

**Experimenting with the model.** When performing simulation studies using discrete event models, multiple measurements of specific quantities *within a single run* are not likely ever to be independent and identically distributed random quantities.<sup>11</sup> Because of this, standard statistical inference approaches for computing confidence intervals and performing hypothesis tests cannot generally be applied to observations made within a single run. However, the results of different runs performed with different streams of pseudo-random data are independent and identically distributed, because they are independently drawn from the same probability distribution by the simulation process.

Thus, to perform a Monte Carlo simulation study whose results can be statistically analyzed, it is necessary to make multiple runs using different streams of pseudo-random data for each run. To ensure that the model represents the ongoing nature of a process, it is generally necessary to run the model for a warmup period so that the simulation starts with a pseudo-random set of initial conditions representative of the actual process. For this study, the warmup period consisted of three months of operative procedure requests to simulate the hospital's backlog of those requests; it also included an additional 12 months of requests and procedures. After the warmup period,

performance measurements were made for a second 12-month period; the simulation was then run for a third 12-month period, to collect waiting time and operative procedure count statistics of patients that arrived during the second 12-month period.

The next step was to compute confidence intervals of the wait times for cardiac operative procedures, the number of cardiac procedures, and the number of cancellations, for each functional ICU capacity value between 6 and 17. It was desired that 99% of the time all of the true means would fall between their computed confidence intervals. To be conservative, it was assumed that the three sets of means were statistically independent. Defining  $P$  to be the likelihood of  $a$  true mean falling between its computed confidence intervals, the likelihood of *all* of the true means falling between their computed confidence intervals equals  $P^{36}$ . This is because the likelihood of 36 independent trials *all* having their true mean falling between their computed confidence intervals, equals the product of each trial having that result, ie,  $P^{36}$ . Setting  $P^{36} = .99$ , the target fraction of the time *all* of the true means fall between their computed confidence intervals, by taking the logarithm of both sides of that equation, it was determined that  $P = .9997209$ . Thus the true mean of each statistic needed to fall within its own confidence intervals 99.97209% of the time. For a sample size of 64 runs per trial, the corresponding  $t$  value is 3.84985, so the confidence intervals were set to the sample means, plus or minus 3.84985 times the standard deviation divided the square root of 64, the number of runs per trial. The sample size of 64 was chosen post priori as it led to narrow confidence intervals.

## RESULTS

An important indicator of ICU performance is the extent that waiting times for requests known to require an ICU stay, from submission of a request until the procedure, are longer than they would be if there was an unlimited number of ICU beds. To measure this, the study measured patient waiting as a function of the hospital's functional ICU capacity. These measurements were performed for patients waiting for cardiac procedures known to require an ICU stay as a proxy for all operative procedures known to require an ICU stay, because cardiac procedures represented the vast majority of the hospital's procedures known to require an ICU stay.

Keeping in mind that the same surgical blocks assignments were used in all of the simulation runs, Table IV shows that as functional ICU capacity decreases, the mean number of days rapidly

increases for patients waiting for cardiac operative procedures. Furthermore, Table V shows that for low levels of functional ICU capacity, not all patients that arrived during the results collection period had their procedures completed before the end of the simulation run. Consequently, the true mean waiting times for patients for low levels of functional ICU capacity are even higher than that displayed in Table IV.

A second indicator of ICU performance is the average number of surgical patients that should be in the ICU, but are parked elsewhere, due to a lack of ICU beds. This phenomenon occurs when patients, after having undergone an operative procedure, require an unexpected ICU stay, either immediately after their procedure, or at some later time. Anecdotally, parking is generally found to be extremely stressful for nurses, and can lead to increased morbidity and mortality for patients needing but not receiving the ICU stay.

Table VI presents the average number of patients requiring an ICU stay that are parked outside the ICU. The 3 pairs of columns on the right each contain an actual ICU capacity, and a column that reports the average number of parked ICU patients for that ICU capacity, for the functional ICU capacity of that row. Thus when the actual ICU capacity is 7 beds and functional ICU capacity is 6, an average of .89 patients needing an ICU stay was parked outside of the ICU. This is equivalent to saying that the average across the different runs of the sum of the hours patients requiring an ICU stay were parked was  $.89 \times 24$  hours per day  $\times 365$  days per year, that is, 7,796 hours per year. Note that while confidence intervals were not computed for these averages, it is clear that parking is a major issue for low levels of functional ICU capacity, unless the total number of ICU beds is at least 1 or 2 beds larger.

A third indicator of ICU performance is the number of operative procedures canceled because of too many ICU beds being occupied. Cancellations are problematic because their occurrence makes it difficult for surgeons to run their practices, because they cannot count on being able to perform operative procedures as planned. It is also problematic because patients tend to react negatively when their operative procedures are canceled.

Table VII presents confidence intervals on the number of blocks in which 1 or more operative procedures were canceled due to a lack of available ICU beds, for each level of functional ICU capacity. This metric was collected instead of canceled operative procedures because of the way operative

**Table IV.** Days wait from time of intervention request through intervention, for different levels of functional ICU capacity

| Functional ICU capacity | Mean days until intervention | Standard deviation of days until intervention | Confidence interval of the mean        |                                        |
|-------------------------|------------------------------|-----------------------------------------------|----------------------------------------|----------------------------------------|
|                         |                              |                                               | Lower limit of days until intervention | Upper limit of days until intervention |
| 6                       | 535                          | 44.89                                         | 513                                    | 556                                    |
| 7                       | 411                          | 42.72                                         | 391                                    | 432                                    |
| 8                       | 304                          | 38.65                                         | 286                                    | 323                                    |
| 9                       | 207                          | 36.88                                         | 189                                    | 225                                    |
| 10                      | 131                          | 28.17                                         | 117                                    | 144                                    |
| 11                      | 79                           | 21.83                                         | 68                                     | 89                                     |
| 12                      | 43                           | 16.91                                         | 35                                     | 52                                     |
| 13                      | 24                           | 11.83                                         | 18                                     | 30                                     |
| 14                      | 13                           | 6.85                                          | 10                                     | 17                                     |
| 15                      | 9                            | 4.65                                          | 7                                      | 12                                     |
| 16                      | 7                            | 3.53                                          | 6                                      | 9                                      |
| 17                      | 7                            | 3.28                                          | 5                                      | 8                                      |

**Table V.** Cardiac operative procedures known to require an ICU stay performed on patients arriving during the 12 months of the results collection period, for different levels of functional ICU capacity

| Functional ICU capacity | Mean number of cardiac interventions | Standard deviation of number of cardiac interventions | Confidence interval of the mean                |                                                |
|-------------------------|--------------------------------------|-------------------------------------------------------|------------------------------------------------|------------------------------------------------|
|                         |                                      |                                                       | Lower limit of number of cardiac interventions | Upper limit of number of cardiac interventions |
| 6                       | 200                                  | 60.7                                                  | 170                                            | 229                                            |
| 7                       | 402                                  | 58.5                                                  | 373                                            | 430                                            |
| 8                       | 582                                  | 41.6                                                  | 562                                            | 602                                            |
| 9                       | 636                                  | 22.2                                                  | 625                                            | 647                                            |
| 10                      | 636                                  | 22.2                                                  | 625                                            | 647                                            |
| 11                      | 636                                  | 22.2                                                  | 625                                            | 647                                            |
| 12                      | 636                                  | 22.2                                                  | 625                                            | 647                                            |
| 13                      | 636                                  | 22.2                                                  | 625                                            | 647                                            |
| 14                      | 636                                  | 22.2                                                  | 625                                            | 647                                            |
| 15                      | 636                                  | 22.2                                                  | 625                                            | 647                                            |
| 16                      | 636                                  | 22.2                                                  | 625                                            | 647                                            |
| 17                      | 636                                  | 22.2                                                  | 625                                            | 647                                            |

procedures are scheduled in the simulation. Nonetheless, the results clearly indicate that the number of cancellations decreases rapidly as the functional ICU capacity is increased.

A fourth ICU performance indicator is the utilization of ICU beds. Ideally, this value should be as high as possible, so long as it does not result in lower values for the other performance indicators. Unfortunately, increasing utilization often does result in lower values for the other performance indicators.

Table VIII presents average ICU utilization, as a function of functional and actual ICU capacity. Interestingly, utilization increases as functional ICU capacity is increased from its lowest value, until at

about 10 beds, utilization starts going down. In addition, for low levels of functional ICU capacity, having a larger number of actual ICU beds increases overall bed utilization.

## DISCUSSION

The goal of this study was to identify potential avenues for improving the performance of the ICU for surgical patients by reducing cancellation of operative procedures known to require ICU stays, reducing wait times for those procedures, and improving access to the ICU. While most ICUs in Canada and in many other parts of the world mix surgical and medical patients, it was felt that an important starting point for an analysis of the need



**Table VI.** Average number of ICU patients parked outside the ICU, for different levels of functional and actual ICU capacity

| <i>Functional ICU capacity</i> | <i>Actual ICU capacity</i> | <i>Average number of parked ICU patients</i> | <i>Actual ICU capacity</i> | <i>Average number of parked ICU patients</i> | <i>Actual ICU capacity</i> | <i>Average number of parked ICU patients</i> |
|--------------------------------|----------------------------|----------------------------------------------|----------------------------|----------------------------------------------|----------------------------|----------------------------------------------|
| 6                              | 6                          | 2.11                                         | 7                          | 0.89                                         | 8                          | 0.28                                         |
| 7                              | 7                          | 1.91                                         | 8                          | 0.75                                         | 9                          | 0.23                                         |
| 8                              | 8                          | 1.75                                         | 9                          | 0.68                                         | 10                         | 0.20                                         |
| 9                              | 9                          | 1.44                                         | 10                         | 0.50                                         | 11                         | 0.13                                         |
| 10                             | 10                         | 1.10                                         | 11                         | 0.36                                         | 12                         | 0.09                                         |
| 11                             | 11                         | 0.83                                         | 12                         | 0.25                                         | 13                         | 0.06                                         |
| 12                             | 12                         | 0.58                                         | 13                         | 0.16                                         | 14                         | 0.03                                         |
| 13                             | 13                         | 0.34                                         | 14                         | 0.10                                         | 15                         | 0.02                                         |
| 14                             | 14                         | 0.18                                         | 15                         | 0.05                                         | 16                         | 0.01                                         |
| 15                             | 15                         | 0.10                                         | 16                         | 0.03                                         | 17                         | 0.01                                         |
| 16                             | 16                         | 0.04                                         | 17                         | 0.01                                         | 18                         | 0.00                                         |
| 17                             | 17                         | 0.02                                         | 18                         | 0.00                                         | 19                         | 0.00                                         |

**Table VII.** Surgical blocks in which there is at least 1 cancellation of an intervention due to a lack of ICU beds, for different levels of functional ICU capacity

| <i>Functional ICU capacity</i> | <i>Mean number of surgical block cancellations</i> | <i>Standard deviation of number of surgical block cancellations</i> | <i>Confidence interval of the mean</i>                       |                                                              |
|--------------------------------|----------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
|                                |                                                    |                                                                     | <i>Lower limit of number of surgical block cancellations</i> | <i>Upper limit of number of surgical block cancellations</i> |
| 6                              | 473                                                | 2.7                                                                 | 472                                                          | 475                                                          |
| 7                              | 404                                                | 2.5                                                                 | 403                                                          | 406                                                          |
| 8                              | 336                                                | 2.3                                                                 | 335                                                          | 337                                                          |
| 9                              | 262                                                | 2.0                                                                 | 261                                                          | 263                                                          |
| 10                             | 195                                                | 1.8                                                                 | 194                                                          | 195                                                          |
| 11                             | 136                                                | 1.5                                                                 | 135                                                          | 137                                                          |
| 12                             | 88                                                 | 1.2                                                                 | 87                                                           | 88                                                           |
| 13                             | 50                                                 | 0.9                                                                 | 49                                                           | 50                                                           |
| 14                             | 24                                                 | 0.6                                                                 | 24                                                           | 25                                                           |
| 15                             | 12                                                 | 0.4                                                                 | 11                                                           | 12                                                           |
| 16                             | 5                                                  | 0.3                                                                 | 5                                                            | 5                                                            |
| 17                             | 2                                                  | 0.2                                                                 | 2                                                            | 2                                                            |

for ICU beds for surgical patients should be with an ICU designed only for surgical patients.

The first step of the study was to develop an analytical tool that could be used to investigate mechanisms for improving the performance of the hospital's ICU. This goal was achieved by building the simulation model discussed in this paper. Perhaps more importantly, a detailed understanding of the underlying processes was developed, and an ability to collect and characterize needed data was also developed. This understanding and ability are currently being used to develop a more detailed second generation simulation model that will be used to simultaneously analyze the overall performance of the operating

theatres, ICU, PACU, and surgical wards. This second generation simulation model is being built so that even with considerably more detail, it will run considerably faster than the current model, and be useful for more applications. It is hoped that this second generation simulation model will lead to scheduling changes that will make it possible for the hospital to improve ICU and surgical service performance, even before new ICU capacity is brought on line.

The second step of the study was to use the tool to determine the factors that affect the ICU's performance, and to develop an understanding of the relationship between those factors and ICU performance. Interestingly, the process of

**Table VIII.** Utilization of ICU beds as a function of actual and functional ICU capacity

| <i>Functional ICU capacity</i> | <i>Total ICU capacity</i> | <i>Utilization of total ICU capacity (%)</i> | <i>Total ICU capacity</i> | <i>Utilization of total ICU capacity (%)</i> | <i>Total ICU capacity</i> | <i>Utilization of total ICU capacity (%)</i> |
|--------------------------------|---------------------------|----------------------------------------------|---------------------------|----------------------------------------------|---------------------------|----------------------------------------------|
| 6                              | 6                         | 57.90                                        | 7                         | 67.10                                        | 8                         | 66.30                                        |
| 7                              | 7                         | 61.40                                        | 8                         | 68.10                                        | 9                         | 66.40                                        |
| 8                              | 8                         | 63.70                                        | 9                         | 68.50                                        | 10                        | 66.40                                        |
| 9                              | 9                         | 66.00                                        | 10                        | 68.80                                        | 11                        | 65.80                                        |
| 10                             | 10                        | 67.20                                        | 11                        | 67.90                                        | 12                        | 64.50                                        |
| 11                             | 11                        | 66.80                                        | 12                        | 66.00                                        | 13                        | 62.40                                        |
| 12                             | 12                        | 65.30                                        | 13                        | 63.50                                        | 14                        | 59.90                                        |
| 13                             | 13                        | 62.50                                        | 14                        | 59.80                                        | 15                        | 56.40                                        |
| 14                             | 14                        | 58.70                                        | 15                        | 55.70                                        | 16                        | 52.50                                        |
| 15                             | 15                        | 54.80                                        | 16                        | 51.80                                        | 17                        | 48.90                                        |
| 16                             | 16                        | 51.50                                        | 17                        | 48.60                                        | 18                        | 46.00                                        |
| 17                             | 17                        | 48.40                                        | 18                        | 45.80                                        | 19                        | 43.40                                        |

determining the performance measures to be analyzed in the study led to an understanding, supported by the results of the study, that actual ICU capacity is not always the appropriate factor for understanding ICU performance. Instead, it is often the case that functional ICU capacity, ie, the minimum level of occupied ICU beds that leads to cancellation of an operative procedure that is known to require an ICU stay, either by itself, or in conjunction with actual ICU capacity, has the strongest impact on performance.

It is important to note that once these steps were performed and results were obtained, hospital management still had to interpret the results, because the study did not yield a single optimal solution. Instead, as indicated in the tables of the results section, the study resulted in a set of tradeoffs, between increased cost and increased performance, that management had to decide between. This process resulted in an immediate increase in staffing from 14–16 actual beds and 13–15 functional beds, to 18–20 actual beds and 17–19 functional beds.

It was also observed, from the results of the study, that utilization of beds in a surgery-only ICU was quite low, even at its highest value. This reinforced the authors' sense that more effort should be made to synchronize demand for ICU beds with their supply; although this realization is not new,<sup>1</sup> the extent of the need for synchronization seems to be underreported. In turn, this suggests the potential benefits of more dynamic scheduling of operative procedures requiring ICU stays. This might possibly be done by delaying the scheduling of operative procedures for a particular range of dates until a point in time much

closer to those dates, so that the current level of utilization of the ICU can be included as a factor in the scheduling.

Although there are differences between the health care systems of different countries, the occurrence of patient flow bottlenecks is universal. Thus perhaps the most important benefit of this study is that it reinforced the authors' sense that simulation is a valuable tool for evaluating proposals for addressing patient flow bottlenecks.

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## DISCUSSION

**Dr Michael S. Nussbaum** (Jacksonville, FL): Thank you for asking me to discuss this paper. Certainly this addresses a complex clinical process that we all face at our institutions, and that's the problem of dealing with bottlenecks. And ICUs in particular are 1 of the primary sites of bottlenecks. Although you addressed obstacles in your manuscript—ie, the operating room, the emergency department—certainly addressing the ICU and the optimal number of beds and capacity is certainly an important aspect. I think utilizing a simulation as you described has a number of advantages. It allows to you create a model that is easier for managers to use and to conceptualize these complex problems. Rather than just asking for more capacity, this really can demonstrate very nicely what drives the need for capacity, and it gives you an opportunity to factor in all of the complex variables. There really are no other analytical methodologies that exist that are really capable of solving some of these challenges. So simulation certainly improves the intuitive feelings and understandings of complex systems as one attempts to define the problem and accomplish the objectives. So my question really runs to some of the disadvantages of these. It sounds like you ran the simulation for 12 months, but prior to that you had several warm-ups. What was the total amount of time and effort required to get to the point of actual running the simulation? What were the costs both in terms of the amount of time required and any specific costs in order to do these simulations and to really carry out this extensive evaluation? My question is, can this approach be easily exported to all of our centers, or does it require this amount of effort at each institution to do this? Or how can one easily translate this into all of our situations? Your results, I think, demonstrate that you do need twice the number of ICU beds than you had at the beginning of this study, so what decisions have been made either by you or by management following this extensive evaluation to arrive to solutions for your bottlenecks? Do you still have 8 ICU beds, or have you increased the number of beds available to you?

**Dr P. Troy** (Montreal, Quebec, Canada): We ran each simulation for a 39-month period. The first 3 months were strictly just to build up accrued requests, because we found that we had approximately a quarter year worth of backlog of requests at the hospital. The next 12 months were just to give us some sense of the system being what's called steady state, that it's actually operating the way it would be in real life. We then ran for 12 months the actual simulation run, plus an additional 12 months to collect data. To give you an idea, to do that 64 times took 15 minutes on a computer. And with the newer version that we are developing right now, it would probably take about a minute, maybe even less. So simulation time is not an issue. Having enough warm-up time is an issue. That's really critically important because you want to make sure that you are

representing reality as much as possible. Now, there are other approaches for determining how long that should be. But we felt that was probably a pretty good indication. With respect to costs, the 2 major pieces of costs had to do with the cost of collecting data and the time for developing the model. But there's also an implicit cost of developing an understanding of the processes and the metrics that needed to be performed and things of that nature. Most hospitals, I'm guessing, have the data they need to do this type of study. For example, most hospitals have a surgical information system on which they track all the operative procedures that they do. I'm not sure whether most hospitals track ICU usage the way our hospital did. I know they have made a special effort over the last several years to collect more data. But I would expect that in the future this type of data would be collected carefully at almost all hospitals. Having said that, once we have done this, particularly if you had access to the model and the methodology, it ought to take a lot less time than it did for us. There's one last piece, and that is a lot of the things that we found that we wanted to measure are things that need to be measured for management purposes anyhow. So, yes, it does cost to build a simulation study. It takes time to get the data. It takes time to understand the process and build the model. But these are things that actually feed into management of the whole operative or surgical services. With respect to answers about the ICU, my understanding is that no changes have yet been made, but there are plans of increasing the size of the ICU, the exact details of which I don't think have been decided.

**Dr Gerald M. Fried** (Montreal, Quebec, Canada): I would like to thank Drs Rosenberg and Troy for providing me their manuscript to review in advance of this meeting. I found it particularly important because of the complexity of this, and I think if I had just looked at the presentation I would have been a bit overwhelmed. It took me some time to actually grasp the concepts expressed and the assumptions made in order to perform this modeling exercise. As I understand it, in performing a modeling process like this, there needs to be some amount of predictability in the surgical requirements for the ICU. For instance, in your environment, you have a large majority of cardiac surgery but not much trauma, for instance, or transplantation or activities in other surgical ICUs that make great demands on the beds. The nice thing about cardiac surgery is that although it may be elective or urgent, it is generally a scheduled activity, and the time of the day, the days of the week, and when those surgeries are performed can be quite predictable. Further, the duration of the ICU after a cardiac surgery event is highly predictable, the duration of the operation again is highly predictable. So I think that these allow you to come up with a model that is quite useful for your particular environment. That leads me to a few questions. You concentrated in this model, as I understand it, on cardiac surgery. But with such a busy emergency room, what proportion of your total ICU demand in your actual hospital was from non-cardiac surgery and

from non-surgical patients, which may be less predictable? If all surgical and non-surgical patients are included, how well did your model predict ICU demand and bed availability? For instance, how did that availability on the floors impact on ICU discharges so that you could discharge patients as you would like to? Also, how would your model work in another real world hospital where you had a large and unpredictable trauma population or an unpredictable but high demand transplant population? Because these populations differ greatly from cardiac surgery in that they present to your ICU in an unpredictable way, often in clusters, they can't be turned away, the surgery when applied is variable in its duration, and the ICU stay associated with care of these patients is quite variable. So as you add one or more of those types of populations to your model, how does it affect the predictability? Finally, have you looked at this type of modeling process to help you determine the relative benefits of having a segregated surgical ICU versus a combined medical surgical ICU?

**Dr P. Troy** (Montreal, Quebec, Canada): With respect to cardiac operative procedures, I would note that, yes, the primary stay, the first stay, is very predictable, but when there are secondary stays they tend to be less predictable, very high variance. When they occur is very variable. And they tend to be longer stays as well. I think the prominent question that you asked is, what will happen in other environments with different mixes? And to get an understanding of that, we need to go back and look at the methodology. First of all, we included all surgeries in the whole simulation. Just in the metrics we only looked at cardiac interventions, and that's the delay time. So we were including all types of operative procedures requiring ICU stays. We were also allowing for emergencies, electives versus nonelectives. And the strength of the methodology that we use is that we look at each intervention type and the relative likelihood of it occurring, and also the relative likelihood of it being an emergency, and we incorporate all those things in the model. So should we go to a different hospital environment, you would take the data, you would characterize it in exactly the same manner, and, yes, you would get very different results. But the process, the methodology that we went through, would not make any difference. It would be done exactly the same way. Now, another really interesting question is whether or not we could use this model—and you didn't ask it quite like this, but I think this was the implied question—could we use this model to predict the availability of an ICU bed tomorrow, the day after, the day after that? We have not done that. But one of our plans is to look at a very similar need and maybe extend the ICU for the post anesthesia care unit to look at when beds will be available and not be available in the PACU based on the interventions that have already taken place on a given day, based on the interventions that are scheduled to take place on a given day, and based on the possible, using simulation, emergency surgery, intervention that is might be required. And all this takes into consideration different possible operative procedure times. So that is

the strength exactly of doing this. Now, what's going to happen, though, when you do this is that you are not going to get a single number. What you are going to get is a distribution of likelihoods that you are going to have one bed free, two beds free, and so on, so you are not going to just get a single result. The last major question that you asked is whether or not you should have a segregated surgical ICU. There has been a lot of research done in this. And basically, to summarize, the result is that there is a one-for-one trade-off. If you have a surgical ICU, what is going to happen is you are going to be able to have more predictability and more control, less cancellations of the operative procedures. However, it's also going to affect the availability of ICU beds for emergency nonoperative procedure requirements. And there's no way to address that. One of the things that we would like to look at for the future is alternative approaches to scheduling that take advantage of the availability of ICU beds right exactly when they occur. And if we can somehow do that, our sense is that we can get better utilization, get more operative procedures done, and also get better overall performance.

**Dr John A. Aucar** (Urbana, IL): Talking about your methodology, what you're using is a combination of probabilistic outcomes here; this is a form of artificial intelligence commonly called "fuzzy logic," and it's applied in a lot of different circumstances. There is an alternative model that uses a combination of discrete outcomes, basically Yes or No switches combined called "neural networks." So my question is, have you considered trying to apply neural network methodology to the same set of data? Because after all, the decision to cancel or not cancel a case based on ICU beds is a discrete dichotomous outcome, either you do or you don't.

**Dr P. Troy** (Montreal, Quebec): Neural networks are essentially a statistical forecasting technique. They are modeled on the way the brain is supposed to work and they use non-linear regression analysis to predict what's going to happen. They come up with a single value. And we did not choose to go this way because you can't come up with a single value for the future as to when there's going to be an elective emergency intervention required or for the exact amount of time an intervention is going to take. We used Monte Carlo simulation in context with discrete event models to represent the events because there's a whole wide variety of things can happen with certain probabilities associated with them. So if we had used neural networks or some other forecasting for prediction technique, we would have lost the value of the probability distribution of the interactions. That's probably the key thing—the interaction between all the different things going on. If you are only looking at one thing, what you suggest makes perfect sense. But when you are looking at interactions, you have to look at all the permutations of those interactions, and Monte Carlo simulation is probably the best way for doing that.

**Dr Richard H. Bell, Jr** (Philadelphia, PA): There are so many processes in hospitals and in health care in general

that involve mismatches of capacity and need, that it seems there's a tremendous need for application of systems and engineering concepts to these things, and yet my sense is that most hospitals really don't have people who do that. So the question I would like to ask you, first of all, personally what your background is and how you got interested in this; and, secondly, whether there is any sort of group or society or discipline that is emerging that can apply systems engineering concepts to health care?

**Dr P. Troy** (Montreal, Quebec, Canada): I have a background in operations research, which for those of you not familiar with it, it's essentially applied math applied to process analysis and improvement. And it's perfectly appropriate for this. There's a lot of math involved, a lot of simulation involved, a lot of probability involved. In terms of societies that address this, first of all, the Mayo Clinic has a conference every year in August where they are looking at issues like this. That's

one. Their Winter Simulation Conference, which is a cross between several different organizations—the Institute for Operations Research Management Science, I think the Association for Computing Machinery, and the IEEE, they have a health care simulation track. There's also a journal, the Machinery Communications of the ACM, which has articles on modeling computer simulation. I think what's happened is in the last several years more and more interest from the operations research and industrial engineering communities is being applied to health care and hospital management. Finally, one of the things that I think needs to happen is that the whole hospitals have to be simulated. Because you have interactions between the emergency departments, surgical services, medical beds, non-medical beds, and all these things interact with each other. We have overflows between different types of beds all the time. We need to start looking at the whole hospital as a system.